

Music-Gen v5 — Cycles 88-90

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Music-Gen v5 — Cycles 88-90

Abstract

Cycles 88-90 are the second through fourth cycles of a new v5 arc — full-length corpus transcription plus rules-extraction under a pre-registered falsification discipline — opened after the v4 campaign closed at c77 and was augmented at c78. The range's science is sound and independently reproducible; its closure at Cycle 90 is not. A detached transcription driver (PID 5201, pinned command, never signalled) landed four of twenty-six focus songs across the range (WIG at c79; CG and PD at c80; Rome at c81 17:04:45Z with note counts drums 1704 / bass 560 / guitar 3909 / other 214 / piano 162 / vocals 438 / full_mix 4439 at 138.52 bars and 151.999 BPM), with Disco A in progress and Essence queued, at approximately 25-27 minutes per song. Cycle 88 landed WIG, the ear v2 test suite, and the v5 flat-band tempo criterion (pre-registered, then falsified). Cycle 89 landed CG and PD, the v5b band-restricted-harmonic-sum tempo criterion (pre-registered, then falsified), a harmony gate record at `n_used=2`, and a `venv-headroom` JSON that the Cycle 90 audit later corrected as measuring free-space-after-build rather than headroom below the 90% FD-1 ceiling. Cycle 90's worker turn ended mid-cycle with a two-line `work_output` fragment; the auditor audited the on-disk state directly. Substantive landings at Cycle 90: reindex hygiene infrastructure (`scripts/v5/harmony_v5.py` `MIDI_DIR_PREFERENCE + MissingReindexError`; new `scripts/v5/reindex_hook.py` with idempotent `reindex_landed`; `scripts/v5/transcribe_full_length.py` calling `post_canonicalize` after manifest write; sidecars for WIG/CG/PD with 7 MIDI SHAs each); the v5c autocorrelation-direct tempo criterion (pre-registered with mtime-proven precedence, 26/26 songs, `byte-det ×2`, verdict `RULES_OUT_AUTOCORR_DIRECT` with `missed_targets=["cdd2717e52820ff6"]` only — PD recovered to 123.047 from 80.75, Rome's 152-vs-103 hemiola resolved, Disco A missed by a 0.0062 margin); reindex fidelity `REINDEX_FIDELITY_DEGRADED` driven solely by the other stem (median `|Δdur|` 0.281s > 0.075 threshold; all other eligible stems hold); groove first data `GROOVE_NONDEGENERATE_STATS_MATCH` on WIG+CG (backbeat 0.1404 corpus vs 0.1355 sampled; bass-kick lock 0.603 vs 0.541); 17 new tests + 51 regression = 68/68; 17/17 FROZEN anchors byte-identical. The auditor's independent recomputation of v5c reproduced the worker's scores to four decimal places and **found the mechanism**: Disco A's parabolic-refined autocorrelation peak sits at lag 21.484 frames = 120.27 BPM — the anchor (120.185) — while the integer candidate at lag 21 reads

its 2T harmonic at lag 42 on the trough side (ac 0.4659) instead of the true 2-beat peak at lag 43 (ac 0.7226); PD's refined peak at 21.146 is near-integer so its 2T at 42 hits the peak (0.7077) and recovers. The v5c miss is a **lag-quantization artifact of a fractional true period, not a wrong periodicity**. Three consecutive pre-registered falsifications (v5 c79, v5b c80, v5c c81) fire the pre-registered meta-gate: the next cycle is a mechanism probe, not a fourth criterion. But the cycle never closed: **zero c81 worker ledger events and zero c81 POR rows** (the emitter tools/_emit_c81_ledger_events.py and tools/_register_c81_por_rows.py exist but were never run), and Rome landed lossy-only under the old driver image so test_reindex_hygiene_c81::test_02 failed live. The auditor applied fix F-1 in place: ran the worker's tested reindex_hook.reindex_landed() on Rome (7 MIDI + 7 .reindexed.json + reindex_manifest.json + sidecar; 15/15 byte-identical across two fresh tempfile.mkdtemp() corpora; 5/5 tests after), recorded ledger event M-V5-CORPUS-1/51e433ade2a845e1-reindexed-c81 (event_id 59579770-0b00-5cc6-83e5-8c174d9d1f2c, agent: auditor, ledger 2000 → 2001), and added the POR row at line 1187. The auditor also corrected two attribution errors: the 156 promise_check ERRORS are not all c23-c70 legacy — lines 1968-2000 (every c78/c79/c80 emitter event) fail missing required field 'agent'; and the c80 venv-headroom advisory used the wrong reference frame, the c81 worker's VENV_BLOCKED_DISK arithmetic was right at 16:51Z but became stale at approximately 17:10Z when disk moved from 86.93% / 5.2 GB to 80% / 7.8 GB (≈4.0 GB consumable below the 90% ceiling; the MANDATORY ear venv is buildable at the next cycle). Independent audit returned **CONTINUE** with one CRITICAL (C-1: closure not landed + Rome lossy-only; Rome half fixed by auditor, ledger/POR half deferred to next cycle) and five MODERATE (ear venv absent third consecutive cycle with blocker now stale; emitter chain since c78 omits required agent field; fidelity statistic confounded by transcription variance; groove model has no bar-phase alignment and near-memorizes; harmony first data unemitted despite gate satisfiable). Canonical 7-key env_pin_sha256=2ac444c3... unchanged. Harness commit messages say "cycle 90" while the ledger says c81; the counter discrepancy is disclosed, not reconciled.

Introduction

The v4 campaign closed at c77 and was augmented at c78. The v5 arc that follows has a different shape: full-length transcription of the 26-song corpus (MuScriptor chunked full-length, canonicalized to MIDI, six stems plus full_mix per song), a reindex layer to repair an index-collision defect inherited from the c22 merge and the c4 serializer, a tempo-detection criterion sought under pre-registered falsification (each candidate criterion is pre-registered with an enum verdict and a meta-gate that fires after three consecutive falsifications), a reindex-fidelity statistic, first-data rules extraction (groove and harmony Markov models), and a MANDATORY ear virtual environment whose build is disk-gated under the FD-1 90% ceiling. The corpus driver runs detached at roughly 25-27 minutes per song and is the binding schedule constraint; disk is the binding resource constraint.

Cycles 88-90 are v5 cycles 2 through 4 (internal c79-c81). The range delivers the reindex infrastructure, three consecutive pre-registered tempo falsifications that converge on a sharp mechanism, and first-data groove statistics — and it fails to land its own closure at Cycle 90 because the worker turn ended mid-cycle before running the emitter. The auditor's role in this range is unusually load-bearing: the on-disk audit substituted for the missing work_output, the auditor patched the Rome lossy-only landing in place, and the auditor's independent recomputation of v5c is what surfaced the lag-quantization mechanism.

Approach

Corpus driver (P0.1). `scripts/v5/transcribe_full_length.py` running detached as PID 5201 under the pinned command, verified alive (`ssl`, 1:55 elapsed at 17:20Z) and never signalled. Landings: WIG (c79), CG (c80), PD (c80), Rome (c81 17:04:45Z), Disco A in progress (other stem done 17:17Z; DONE $\approx$ 17:30Z), Essence next. `df` at Cycle 90 open 86.93% under user semantics (5.2 GB avail); at approximately 17:10Z disk changed to 80% / 7.8 GB avail — the driver's [`df`] WARN lines stop before `muscriptor:guitar` at 17:10; the cause is not attributable from logs. `/tmp` (5.7 GB on the same ext4 volume `/dev/vda`) holds $\approx 45 \times 70$ MB `c37b_v4_s*_r*` render transients (200 h old) plus 0.28 GB `tfhub_modules` (227 h) — ≈ 3.3 GB reclaimable.

Reindex hygiene (P0). The index-collision defect (c22 merge + c4 serializer) produces lossy canonical MIDI where chunk-local note indices collide across chunks. Cycle 90 landed: `scripts/v5/harmony_v5.py` (sha 7a41fff0...) with `MIDI_DIR_PREFERENCE = ("canonical_v5c_reindexed", "canonical_v5_reindexed")` at L78 and `MissingReindexError("MISSING_REINDEX: ...")` at L177 so consumers refuse lossy directories; new `scripts/v5/reindex_hook.py` with `post_canonicalize`, `write_sidecar`, and idempotent `reindex_landed`; `scripts/v5/transcribe_full_length.py` (sha c1c6b2df...) importing the hook at L70 and calling `post_canonicalize` after the manifest write at L309. Sidecars present for WIG/CG/PD with 7 MIDI SHAs + `reindex_manifest` SHA; c79/c80 manifest bytes untouched (WIG manifest sha fc571725... recorded in sidecar). `tempo_v5b_summary_c81.tsv` carries `anchor_source`; WIG anchor 99.384 (`librosa_full_mix_c79`).

Ear venv (P1). `data/v5/ear/venv_build_c81.json` records `VENV_BLOCKED_DISK` at 16:51Z: the brief gate (≥ 3.5 GB free) passed, but under the driver's `used/(used+avail)` semantics only 1.222 GB was consumable below the 90% FD-1 ceiling versus a 2.36 GB build (post-build 92.86%). No `pip`, no SIGSTOP, PID untouched, `main-env pip freeze sha 90ed1d9f... == c79`, one-line operator request (`/root/.local`, 2.97 GB) present, `scripts/v5/ear_probe_v5.py` exits 3 `EAR_VENV_ABSENT`.

Tempo v5c (P2). Pre-registered autocorrelation-direct criterion. Pre-registration mtime 1788713418.14 precedes the earliest `tempo_v5c.json` at 1788713479.38 (26/26; `preregistration_gate.prereg_precedes_all_outputs=true`). Env pin unchanged. Byte-det $\times 2$ 26/26. Figure `fig_tempo_v5c_scores.png` with co-located `plot_tempo_v5c_scores.py`; caption viewed and accurate. Non-anchor flips 6/21 = 28.6% ($< 50\%$ pre-declared bound).

Reindex fidelity (P3). `reindex_fidelity_c81.json` comparing reindexed full-length MIDI against the 30 s section transcriptions from c20 on pitch-matched onsets. Byte-det $\times 2$ holds (`c81_p3_p4_byte_determinism.json`).

Rules first data (P4). `groove_v5.json` on WIG + CG (172 bars) with a pre-declared degeneracy enum. Harmony: `harmony_v5_gated.json` remains the c80 record (`n_used` 2); no c81 emission.

Tests / anchors / validators (P5). 17 new tests (`test_reindex_hygiene_c81` 5, `test_tempo_v5c` 6, `test_groove_v5` 3, `test_ear_venv_c81` 3) + 51 regression = 68/68. FROZEN anchors 14/14 + 3 extras byte-identical. `/usr/bin/python3 guard`, no PRNG / `sidecar_nonfactor` / VST3 state APIs on the 6 new scripts (`test_tempo_v5c::test_06`). `promise_check` 156 ERROR / 7260 WARN, rc 1; `org_check` 0 ERROR / 54 WARN.

Structured-state landing. `tools/_emit_c81_ledger_events.py` (309 lines, ~ 17 events, reads landed/sided state from disk at emit time, tolerates foreign-scratchpad inputs being absent, omits agent) and `tools/_register_c81_por_rows.py` exist but were never run. Ledger at 2000 lines: 0 c81 worker events; POR: 0 c81 rows.

Findings

Tempo: three consecutive falsifications converge on a mechanism

v5c verdict RULES_OUT_AUTOCORR_DIRECT, missed_targets=["cdd2717e52820ff6"] (Disco A) only. Hits: WIG 99.384 ($\Delta 0$), CG 92.285 ($\Delta +1.559$), Rome 152.00 ($\Delta 0$; hemiola table present with $s(\text{lag}17)=1.401$ vs $s(\text{lag}25)=1.112$), PD 123.047 ($\Delta 0$ — recovered from 80.75 under v5b). Disco A: candidate 80.75 scores 1.1913 vs candidate 123.05 scores 1.1851 — a margin of 0.0062.

The auditor's independent recomputation (librosa 0.11, same parameters) reproduces the worker's scores to four decimal places: Disco A $s(21)=1.1851$, $s(32)=1.1913$; PD $s(21)=1.3788$, $s(32)=1.2223$. The mechanism: Disco A's parabolic-refined autocorrelation peak is at lag 21.484 frames = 120.27 BPM — the anchor is 120.185. The integer candidate at lag 21 evaluates its 2T harmonic at lag 42, which sits on the trough side (ac 0.4659), while the true 2-beat peak is at lag 43 (ac 0.7226; refined 43.04). PD's refined peak is 21.146 — close enough to integer that its 2T at 42 lands on the peak (0.7077), which is why PD recovered. Reading Disco A's 2T at $2 \times 21.484 = 42.97$ gives $ac \approx 0.72$ and $s \approx 1.31 > 1.19$. The v5c miss is a lag-quantization artifact of a fractional true period, not a wrong periodicity.

Three consecutive pre-registered falsifications — v5 flat-band (c79), v5b band-restricted harmonic sum (c80), v5c autocorrelation-direct (c81) — fire the pre-registered meta-gate. The next cycle is a mechanism probe with no fourth criterion. Each falsification isolated a sharper mechanism (candidate-band zeroing → integer-lag quantization); the worker did not retune, and FD-1 held. `recanonicalization_blocked.json` is unchanged: PD and Disco A remain blocked for rules, which is correct under RULES_OUT even though PD hit.

Reindex hygiene: infrastructure landed; Rome landed lossy and was patched by the auditor

The consumer-side refusal (`MissingReindexError`), the hook, the driver integration, and the sidecars for WIG/CG/PD all landed and are tested (`test_01` confirms a lossy-only fixture raises). But Rome landed at 17:04:45Z under the old driver image — the running PID 5201 predates the hook integration — so Rome arrived lossy-only and `test_02` failed live with missing sidecar for 51e433ade2a845e1.

Fix F-1 (auditor, in place): ran the worker's tested `scripts.v5.reindex_hook.reindex_landed()` on Rome, producing `canonical_v5_reindexed/` (7 MIDI + 7 `.reindexed.json` + `reindex_manifest.json`) plus the sidecar. Verification: reindexed MIDI note_on counts equal JSON starts 7/7; two fresh `tempfile.mkdtemp()` corpora with inputs copied yield 15/15 files byte-identical to in-place; `test_reindex_hygiene_c81.py` 5/5; harmony smoke parses Rome. Recorded as ledger event M-V5-CORPUS-1/51e433ade2a845e1-reindexed-c81 (event_id 59579770-0b00-5cc6-83e5-8c174d9d1f2c, agent: auditor, validated, supersedes_path: null; ledger 2000 → 2001) plus a POR row at line 1187 inside `## Milestones`. The auditor's first append lacked agent (inherited the c78+ template) and raised ERROR 157; the auditor corrected its own uncommitted line in place with a new UUID5, restoring `promise_check` to 156 with line 2001 clean.

New MINOR finding from F-1: Rome bass has 122/560 unpaired starts, all at $t = 215.08$ s — a tail chunk with 120 simultaneous starts whose same-index ends precede them — disclosed with 100 ms synthetic durations. Other stems have 0 unpaired (53 bass ends with span > 30 s are dedup'd-start leftovers). This is an input artifact, not a pairing failure.

Disco A will land lossy-only at $\approx 17:30$ Z under the same old image and needs the same catch-up; every later landing does too until the driver is restarted.

Ear venv: honest blocker at 16:51Z, stale by 17:10Z

The c81 worker's VENV_BLOCKED_DISK arithmetic was right at 16:51Z: 1.222 GB consumable below the 90% ceiling versus a 2.36 GB build. The auditor corrects its own c80 audit here: `venv_headroom_c80.json`'s `build_allowed_if_sequenced_after_transcription=true` measured free-space-after-build (2.61 GB), not headroom below the ceiling. But at 17:13Z+ the arithmetic flips: 30 GB used / 7.8 GB avail $\rightarrow \approx 4.0$ GB consumable $\rightarrow$ post-build (2.36 GB + 0.28 GB VGGish) $\approx 86.3\% < 90\%$. The MANDATORY venv is buildable at the next cycle. This is the third consecutive cycle without it.

Reindex fidelity: verdict honest, statistic confounded

REINDEX_FIDELITY_DEGRADED, driven solely by other (median $|\Delta_{dur}|$ 0.281 s > 0.075 on 305 pitch-matched onsets; F1 0.746; pairing-ambiguity 0.90). All other eligible stems hold: drums 0.1 ms (F1 0.79 any-pitch / 0.55 pitch), piano 18 ms (F1 0.44), vocals 20 ms (F1 0.62), full_mix 2 ms (F1 0.53); bass 1.5 ms (F1 0.95 but 19 $<$ 20 pitch-matched $\rightarrow$ ineligible); guitar 0 notes in the c20 section window. Caveat: statistic (b) compares two different MuScriptor runs (30 s section serialized at 50.174 BPM versus chunked full-length), so $|\Delta_{duration}|$ mixes transcription variance with pairing error. The two-chunk fixture (`test_03`) proves count-losslessness, not duration correctness under cross-chunk index collisions. The other-stem DEGRADED verdict cannot yet be attributed to the greedy pairing.

Rules first data: groove passes its pre-declared enum; harmony unemitted

`groove_v5.json` = GROOVE_NONDEGENERATE_STATS_MATCH on WIG + CG (172 bars): backbeat 0.1404 corpus vs 0.1355 sampled ($\Delta 0.005$); bass-kick lock 0.603 vs 0.541 ($\Delta 0.062$); 64 sampled bars, 51 distinct kick / 38 distinct bass patterns; context counts 124/148/124; PD + Disco A refused. The audit's M-4 caveat: the bar grid is anchored at $t=0$ with no downbeat/phase alignment (`groove_v5.py` has no phase/offset handling), and 124 distinct kick patterns over 172 bars at ~ 1 count per context is near-memorization — the in-sample STATS_MATCH is close to tautological. Honest under the pre-declared enum, not yet generator-grade.

Harmony: `harmony_v5_gated.json` is still the c80 record (n_{used} 2); no `harmony_v5-gated-c81` emission although the brief asked for one. After F-1 the ≥ 3 unblocked-song gate is satisfiable (WIG, CG, Rome). In-memory smoke (no artifact written): `analyse_song(Rome)` $\rightarrow$ B major (KK corr 0.791), 557 beats, 262 segments; WIG G major 0.924 unchanged from c80.

Validators: attribution corrected

`promise_check` 156 ERROR / 7260 WARN. The 156 are not all c23-c70 legacy: lines 1968-2000 — every c78/c79/c80 emitter event — each fail missing required field 'agent' (REQUIRED_EVENT_FIELDS includes `agent` $\in$ canonical assessors). The c80 audit's "0 c80-attributable" claim and the c81 worker's baseline note were wrong on attribution; counts were right. 33 of the 156 are c78-c80-attributable; the unrun c81 emitter would add ~ 17 more. WARNs +72 vs the worker's 7188 baseline = unadopted c81 artifacts under `data/v5/` (no c81 events landed). `org_check` 0 ERROR / 54 WARN unchanged.

Anchors and tests

FROZEN anchors 14/14 + 3 extras byte-identical: `cert` a6876911..., `rules_artifact` e19fb205..., `cg_ab_mix` 6e13e007..., `bass_v2` 2a1cb340..., `PD stem_manifest` d483f2bf..., `exemplar_set` 31c10dfb..., `v4_ear` e775621b..., `iterate_v4` 8f1f0b88..., `interpolate_v4` 2359f35d..., `report_v3` b900b0ee..., `recreate_v3` b1490874..., `stage_cache` 33435a84..., `serializer` bbf015f..., `hygiene` c27 771ff42b...; plus `recreate_v3_`

checkpointed da418986..., launch_detached 999045f3..., corpus_manifest 73362136.... Tests 68/68 (17 new after F-1 + 51 regression: tempo_v5b 6, harmony_v5 3, tempo_v5 7, gen_interpolate 6, ear_v2 9, ear_batch 8, ear_scaffold 5, gen_iterate 7).

Audit outcome

CONTINUE. One CRITICAL, five MODERATE.

- **C-1 (CRITICAL)** — c81 closure not landed + Rome lossy-only on disk (live test failure; S1/S6 broken). Rome half FIXED by auditor (F-1); ledger/POR half deferred to next cycle.
- **M-1** — Ear venv absent for the third consecutive cycle; the c81 blocker was correct at 16:51Z but stale since ~17:10Z (≈ 4.0 GB consumable); c80 headroom JSON used the wrong reference (auditor miss, corrected).
- **M-2** — Emitter chain since c78 omits the required agent field $\rightarrow$ 33 of 156 ERRORS are c78-c80-attributable; the unrun c81 emitter would add ~17 more.
- **M-3** — Fidelity statistic (b) confounded by transcription variance; other-stem DEGRADED cannot yet be attributed to the greedy pairing.
- **M-4** — Groove model: bar grid at $t=0$ with no phase alignment; near-memorization; in-sample STATS_MATCH close to tautological.
- **M-5** — Harmony first data not emitted although the brief asked for it; gate now satisfiable.

Sufficiency: S1 (no consumer path to lossy MIDI) MET only after F-1; S2 (venv or honest block) MET via the blocked branch but the blocker no longer holds; S3 (tempo prereg, 26/26, enum, byte-det, Rome table) MET; S4 (fidelity JSON) MET with the confound caveat; S5 (harmony emitted or gated; groove verdict) PARTIAL; S6 (tests, regression, promise_check, ledger + POR, anchors) NOT MET — zero worker ledger events / POR rows.

Rationale: the empirical work is sound and independently reproducible (v5c to 4 dp; byte-det 28/28; anchors 17/17; 68/68 tests), and the pre-registered falsification discipline held. But the cycle never closed. None of the gaps is a science defect; they are landing gaps the next cycle can close in its first hour before advancing the mechanism probe. No invalidated event is warranted: the c80 headroom advisory was wrong in reference frame, but the c80 closure verdicts stand.

Discussion

Three things about this range are worth naming.

First, the auditor's independent recomputation is what turned a third falsification into a mechanism. The worker's v5c run was correct, honest, and byte-deterministic; it reported RULES_OUT_AUTOCORR_DIRECT with Disco A missed by 0.0062 and did not retune. Under the meta-gate that was enough to stop the criterion search. But the audit did not stop at verifying the numbers — it recomputed them to four decimal places, then looked at the parabolic-refined peaks, and found that Disco A's true period is 21.484 frames, that the integer candidate at 21 evaluates its 2T harmonic one frame short of the real 2-beat peak, and that PD escapes only because its refined period happens to be near-integer. That is a specific, testable mechanism (integer-lag quantization of a fractional true period causing off-peak harmonic reads) with a specific confirmation statistic (refined-lag BPM within ± 1 of all five anchors; Disco A's harmonic sum at refined $T/2$ and $2T$ exceeding the 80.75 candidate's). The range's value is not that v5c failed; it is that the failure is now explained well enough that the next cycle can test the explanation rather than guess a fourth criterion.

Second, the range's closure failure is a process defect that the on-disk-authoritative discipline

was designed to survive, and it did — at real cost. The worker turn ended mid-cycle with a two-line fragment; the emitter and POR registration scripts were written, tested, and never run; Rome landed lossy under an old driver image and a live test failed. The audit substituted a direct on-disk verification for the missing work_output, patched Rome in place with the worker’s own tested hook (proving byte-identity across two fresh tempdirs before recording it), recorded the fix with agent: auditor so the provenance is honest, and caught its own inherited omission of the agent field on the first append. This is the right shape for an auditor fix — minimal, tested, attributed, and disclosed — but it inverts the intended role split, and the ledger/POR half of C-1 remains open. The next cycle’s first act is to run the emitter with agent added and to skip the Rome event the auditor already landed.

Third, the agent field omission is a recurring process defect that survived three cycles because attribution was assumed rather than checked. The c78+ emitter template dropped a required field; the c80 audit claimed “0 c80-attributable” ERRORS; the c81 worker’s baseline note repeated it; both were wrong on attribution while right on counts. The audit’s correction — check validator attribution by line number, not by assumption — is the generalizable lesson, and the recommended _infra/emitter-agent-field-c82 disclosure event (do not rewrite history) is the right remedy. The same pattern shows up in the venv-headroom JSON: a correct-looking number measured against the wrong reference frame, propagated one cycle before being caught. Both are the kind of drift that a pre-registered, on-disk-authoritative discipline eventually catches — but “eventually” cost a cycle each.

Open questions

- **Close c81 (first act, ≤20 min).** Add "agent": "worker" to every event body in tools/_emit_c81_ledger_events.py and every future emitter; disclose the 33 c78-c80 failures via a _infra/emitter-agent-field-c82 event without rewriting history; run the emitter (skipping M-V5-CORPUS-1/51e433ade2a845e1-reindexed-c81, which exists with agent: auditor and a POR row at line 1187); run tools/_register_c81_por_rows.py; re-pin harmony_v5.py / transcribe_full_length.py SHAs in the events; run promise_check and report attribution honestly (expect 156 baseline + 0 new).
- **Reindex catch-up.** Run PYTHONPATH=. /usr/bin/python3 scripts/v5/reindex_hook.py after Disco A’s manifest lands (≈17:30Z) and after every later landing until the driver is restarted. Recommend restarting PID 5201 at a song boundary (stage cache skips finished stages) via READ-ONLY launch_detached with the pinned command so the hook is live at birth.
- **Ear venv, MANDATORY, now feasible.** Pre-register the ceiling arithmetic in the driver’s used/(used+avail) semantics (≈79% now; ≈4.0 GB consumable; expected post-build ≈86.3%). Optionally reclaim ≈3.3 GB first by deleting the 200 h-old /tmp/c37b_v4_s*_r* and /tmp/tfhub_modules transients via Python (not deliverables; record a hygiene event). Then kill -STOP the driver tree at a stage boundary, build with the c79 pinned command, run ear_probe_v5.py ×2 vs cached rows (enum EAR_VENV_REPRODUCES_CACHE / _DIFFERS_FROM_CACHE), write env_pin_ear_venv_c81.json receipts, verify main-env freeze sha 90ed1d9f..., kill -CONT; abort/remove only if df ≥ 90%.
- **Harmony first data.** Gate met (WIG, CG, Rome). Run harmony_v5.py → harmony_markov_v5.json with the c80 pre-declared degeneracy verdict; disclose the Rome bass tail artifact (120 notes at 215.08 s, 100 ms synthetic) and consider excluding beats with anomalous polyphony.
- **Tempo — meta-gate fired: mechanism probe only.** Pre-declare per-song normalized-autocorrelation dumps; parabolic peak refinement of every local maximum; drums-stem vs full-mix librosa.beat.beat_track comparison. Candidate mechanism: integer-lag quantization of fractional true periods causes off-peak harmonic reads (Disco A refined 21.484

→ 120.27 BPM; $ac(42)=0.466$ vs $ac(43)=0.723$; PD refined 21.146 unaffected). Confirmation statistic (a measurement, not a pick rule): refined-lag BPM within ± 1 of all five anchors, and Disco A's harmonic sum at refined T/2 and 2T exceeding the 80.75 candidate's. Any subsequent criterion is a c83+ decision with its own pre-registration.

- **Groove #2 before any generator use.** Pre-declared bar-phase alignment (offset maximizing kick-on-1 + snare-on-2/4 mass); a held-out song for validation (Rome now available); a coarser pattern alphabet or additive smoothing so contexts are not singletons.
- **Fidelity follow-up (M-3).** A synthetic fixture with the same chunk-local index in two chunks within 30 s and known durations (assert durations, not just counts); if `stage_cache` retains per-chunk MuScriptor JSON, re-pair within chunk as ground truth for other.
- **Carry.** PD/Disco A remain blocked for rules; corpus ETA $\approx$ two cycles out; `stall_counter.json` 0/12 untouched.
- **Cycle-counter discrepancy.** Harness commit messages say “cycle 90”; the ledger says c81. Disclosed, not reconciled.
- **MINOR log (not acted on).** Prereg `registered_utc` 16:52:00 vs file `mtime` 16:50:18; `harmony_v5.py` header still says `cycle: 80`; emitter hardcodes TS 17:30:00Z and a foreign session scratchpad path (guarded); `test_tempo_v5c` prints “6/6” while the brief counted differently; +72 orphan WARNs on `data/v5/**`; `transcription_progress.json` still self-healing dry-run; Rome bass 215.08 s tail chord; `tools/_c25_poll_result.log` and `tools/logs/` untracked; `shutil` vs `df` percentage skew (known).

Appendix: Provenance

Directive. Execute the Music-Gen v4 closure campaign (closed c77, augmented c78); the range covers v5 cycles 2-4 of the successor corpus-transcription + rules-extraction arc.

Cycle range. cycles 88–90 (internal c79–c81; harness commit messages say “cycle 90” while the ledger says c81 — disclosed, not reconciled).

Working directory. `/home/user/long-exposure-runs/music-gen.`

Session references.

- Cycle 88 researcher `3eed8c3a-bb20-4abe-ba0e-464259fad0ff`; worker `e8b1bb81-3dbf-4dee-af8d-ed739d07f97`; auditor `d7421dc4-f4e2-4401-b232-34076645faaf`.
- Cycle 89 researcher `82e6f87d-0a12-4dc4-b899-942ba0b2904b`; worker `1039f5c7-ae74-4f76-9fa8-807fe3a42c67`; auditor `2c96aafa-193c-4775-bd8d-649310aadbad`.
- Cycle 90 researcher `7212c597-3f31-48ae-8e6a-5314b62a8850`; worker `8913f428-c77c-4aee-b9ba-44cb6170dfd3`; auditor `c18ac1c0-97d8-4e8c-9ebd-fae30dfad738`.

Audit verdict. CONTINUE. One CRITICAL (C-1: c81 closure not landed + Rome lossy-only; Rome half fixed by auditor F-1, ledger/POR half deferred). Five MODERATE (M-1 ear venv absent third consecutive cycle with stale blocker; M-2 emitter chain since c78 omits required agent field; M-3 fidelity statistic confounded; M-4 groove model no phase alignment / near-memorization; M-5 harmony first data unemitted despite satisfiable gate). One MINOR from F-1 (Rome bass 122/560 unpaired starts at $t=215.08$ s, input artifact). On-disk state verified live 2026-09-06 16:47–17:27Z; the c81 `work_output` was a two-line fragment.

Corpus driver. PID 5201, pinned command, never signalled. Landed: WIG (c79), CG (c80), PD (c80), Rome (c81 17:04:45Z; drums 1704 / bass 560 / guitar 3909 / other 214 / piano 162 / vocals 438 / full_mix 4439; 138.52 bars at 151.999 BPM; `df_max` 86.9%). Disco A in progress (DONE $\approx$ 17:30Z), then Essence. 4/26 landed; $\approx$ 25–27 min/song. Disk moved 86.93% / 5.2 GB → 80% / 7.8 GB at $\approx$ 17:10Z (cause not attributable from logs). $\approx$ 3.3 GB reclaimable in `/tmp` (c37b_v4_s*_r* transients 200 h; `tfhub_modules` 227 h).

Substantive landings this range.

- Reindex hygiene: scripts/v5/harmony_v5.py (sha 7a41fff0...) MIDI_DIR_PREFERENCE L78 + MissingReindexError L177; scripts/v5/reindex_hook.py (new; post_canonicalize, write_sidecar, idempotent reindex_landed); scripts/v5/transcribe_full_length.py (sha c1c6b2df...) hook import L70, post_canonicalize call L309; sidecars for WIG/CG/PD (7 MIDI SHAs + reindex_manifest SHA each; WIG manifest sha fc571725...); tempo_v5b_summary_c81.tsv with anchor_source.
- Tempo v5c: RULES_OUT_AUTOCORR_DIRECT; prereg mtime 1788713418.14 < earliest output 1788713479.38; 26/26; byte-det ×2 (tempo_v5c_byte_determinism.json); missed_targets=["cdd2717e52820ff6"]; hits WIG 99.384 / CG 92.285 / Rome 152.00 (hemiola s(17)=1.401 vs s(25)=1.112) / PD 123.047; Disco A 80.75 s=1.1913 vs 123.05 s=1.1851; non-anchor flips 6/21; fig_tempo_v5c_scores.png + plot_tempo_v5c_scores.py. Auditor recomputation to 4 dp; mechanism: Disco A refined peak 21.484 = 120.27 BPM; ac(42)=0.4659 vs ac(43)=0.7226; PD refined 21.146. recanonicalization_blocked.json unchanged.
- Reindex fidelity: reindex_fidelity_c81.json = REINDEX_FIDELITY_DEGRADED (other: median |Δdur| 0.281 s, F1 0.746, ambiguity 0.90, 305 onsets; drums 0.1 ms F1 0.79/0.55; piano 18 ms F1 0.44; vocals 20 ms F1 0.62; full_mix 2 ms F1 0.53; bass 1.5 ms F1 0.95 but 19 < 20 ineligible; guitar 0 notes in window). Byte-det ×2 (c81_p3_p4_byte_determinism.json).
- Groove: groove_v5.json = GROOVE_NONDEGENERATE_STATS_MATCH (WIG + CG, 172 bars; backbeat 0.1404 vs 0.1355; bass-kick lock 0.603 vs 0.541; 64 sampled bars; 51 kick / 38 bass patterns; contexts 124/148/124).
- Ear venv: data/v5/ear/venv_build_c81.json = VENV_BLOCKED_DISK (1.222 GB consumable vs 2.36 GB build at 16:51Z); ear_probe_c81.json exit 3 EAR_VENV_ABSENT; main-env freeze sha 90ed1d9f... == c79.
- Auditor fix F-1: Rome canonical_v5_reindexed/ (7 MIDI + 7 .reindexed.json + reindex_manifest.json) + sidecar; 7/7 note_on == starts; 15/15 byte-identical across two fresh tempdirs; ledger event M-V5-CORPUS-1/51e433ade2a845e1-reindexed-c81 (event_id 59579770-0b00-5cc6-83e5-8c174d9d1f2c, agent: auditor, supersedes_path: null); POR row line 1187; ledger 2000 → 2001.

Structured-state landing. 0 c81 worker ledger events; 0 c81 POR rows. tools/_emit_c81_ledger_events.py (309 lines, ~17 events, omits agent) and tools/_register_c81_por_rows.py exist, never run.

Tests. 68/68: new test_reindex_hygiene_c81 5/5 (after F-1), test_tempo_v5c 6/6, test_groove_v5 3/3, test_ear_venv_c81 3/3; regression 51/51 (tempo_v5b 6, harmony_v5 3, tempo_v5 7, gen_interpolate 6, ear_v2 9, ear_batch 8, ear_scaffold 5, gen_iterate 7).

FROZEN anchors (14/14 + 3 extras byte-identical). cert a6876911...; rules_artifact e19fb205...; cg_ab_mix 6e13e007...; bass_v2 2a1cb340...; PD stem_manifest d483f2bf...; exemplar_set 31c10dfb...; v4_ear e775621b...; iterate_v4 8f1f0b88...; interpolate_v4 2359f35d...; report v3 b900b0ee...; recreate_v3 b1490874...; stage_cache 33435a84...; serializer bbf015f...; hygiene c27 771ff42b...; recreate_v3_checkpointed da418986...; launch_detached 999045f3...; corpus_manifest 73362136...

Validators. promise_check 156 ERROR / 7260 WARN (rc 1): lines 1968-2000 (all c78/c79/c80 emitter events) fail missing required field 'agent' — 33 of 156 c78-c80-attributable; +72 WARN = unadopted c81 data/v5/ artifacts. org_check 0 ERROR / 54 WARN.

Environment pin. Canonical 7-key env_pin_sha256=2ac444c3... unchanged. /usr/bin/python3 guard, no PRNG / sidecar_nonfactor / VST3 state APIs on the 6 new scripts.

v5 arc status at range close. Cycle 3 of ~6. Corpus 4/26 landed, all lossless after F-1; Disco A ≈17:30Z. Tempo: three pre-registered criteria falsified; meta-gate fired; next cycle

is a mechanism probe. Index-collision defect contained (hook in driver, consumers refuse lossy dirs, catch-up loop idempotent); FROZEN 30 s deliveries unaffected. Ear venv absent (buildable next cycle). Groove first data landed (not generator-grade). Harmony first data gate satisfiable, unemitted. `stall_counter.json` 0/12 untouched.